

CHAPTER -3 LOOKING INSIDE LARGE LA

```
# install these things
!pip install -U \
transformers==4.46.3 \
huggingface_hub==0.26.2 \
accelerate==1.1.1 \
tokenizers==0.20.3 \
safetensors
```

```
Requirement already satisfied: transformers==4.46.3 in /usr/local/lib/python3
Requirement already satisfied: huggingface_hub==0.26.2 in /usr/local/lib/pyth
Requirement already satisfied: accelerate==1.1.1 in /usr/local/lib/python3.12
Requirement already satisfied: tokenizers==0.20.3 in /usr/local/lib/python3.1
Requirement already satisfied: safetensors in /usr/local/lib/python3.12/dist-
Requirement already satisfied: filelock in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: numpy>=1.17 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: packaging>=20.0 in /usr/local/lib/python3.12/d
Requirement already satisfied: pyyaml>=5.1 in /usr/local/lib/python3.12/dist-
Requirement already satisfied: regex!=2019.12.17 in /usr/local/lib/python3.12
Requirement already satisfied: requests in /usr/local/lib/python3.12/dist-pac
Requirement already satisfied: tqdm>=4.27 in /usr/local/lib/python3.12/dist-p
Requirement already satisfied: fsspec>=2023.5.0 in /usr/local/lib/python3.12/
Requirement already satisfied: typing-extensions>=3.7.4.3 in /usr/local/lib/p
Requirement already satisfied: psutil in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: torch>=1.10.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: setuptools<82 in /usr/local/lib/python3.12/dis
Requirement already satisfied: sympy>=1.13.3 in /usr/local/lib/python3.12/dis
Requirement already satisfied: networkx>=2.5.1 in /usr/local/lib/python3.12/d
Requirement already satisfied: jinja2 in /usr/local/lib/python3.12/dist-packa
Requirement already satisfied: cuda-toolkit==12.8.1 in /usr/local/lib/python3
Requirement already satisfied: cuda-bindings<13,>=12.9.4 in /usr/local/lib/py
Requirement already satisfied: nvidia-cudnn-cu12==9.19.0.56 in /usr/local/lib
Requirement already satisfied: nvidia-cusparselt-cu12==0.7.1 in /usr/local/li
Requirement already satisfied: nvidia-nccl-cu12==2.28.9 in /usr/local/lib/pyt
Requirement already satisfied: nvidia-nvshmem-cu12==3.4.5 in /usr/local/lib/p
Requirement already satisfied: triton==3.6.0 in /usr/local/lib/python3.12/dis
Requirement already satisfied: nvidia-cublas-cu12==12.8.4.1.* in /usr/local/l
Requirement already satisfied: nvidia-cuda-runtime-cu12==12.8.90.* in /usr/lo
Requirement already satisfied: nvidia-cufft-cu12==11.3.3.83.* in /usr/local/l
Requirement already satisfied: nvidia-cufile-cu12==1.13.1.3.* in /usr/local/l
Requirement already satisfied: nvidia-cuda-cupti-cu12==12.8.90.* in /usr/loca
Requirement already satisfied: nvidia-curand-cu12==10.3.9.90.* in /usr/local/
Requirement already satisfied: nvidia-cusolver-cu12==11.7.3.90.* in /usr/loca
Requirement already satisfied: nvidia-cusparse-cu12==12.5.8.93.* in /usr/loca
Requirement already satisfied: nvidia-nvjitlink-cu12==12.8.93.* in /usr/local
Requirement already satisfied: nvidia-cuda-nvrtc-cu12==12.8.93.* in /usr/loca
Requirement already satisfied: nvidia-nvtx-cu12==12.8.90.* in /usr/local/lib/
Requirement already satisfied: charset_normalizer<4,>=2 in /usr/local/lib/pyt
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.1
Requirement already satisfied: certifi>=2017.4.17 in /usr/local/lib/python3.1
Requirement already satisfied: cuda-pathfinder~=1.1 in /usr/local/lib/python3
Requirement already satisfied: mpmath<1.4,>=1.1.0 in /usr/local/lib/python3.1
Requirement already satisfied: MarkupSafe>=2.0 in /usr/local/lib/python3.12/d
```

```
# remove any partially downloaded files are there
!rm -rf ~/.cache/huggingface/hub/models--microsoft--Phi-3-mini-4k-instruct
```

```
#torch is the main library of PyTorch, a popular deep learning framework.
#It is used to create, train, and run neural networks, including LLMs
import torch

# import the model , tokenizer and pipeline from transformers library
from transformers import AutoModelForCausalLM, AutoTokenizer, pipeline

# Load model and tokenizer
tokenizer = AutoTokenizer.from_pretrained("microsoft/Phi-3-mini-4k-instruct")
model = AutoModelForCausalLM.from_pretrained(
    "microsoft/Phi-3-mini-4k-instruct",
    device_map="auto",
    torch_dtype="auto",
    trust_remote_code=True,)

# Create a pipeline (interface to work with models)
generator = pipeline(
    # Tells to generate the text (task type)
    "text-generation",
    model=model,
    tokenizer=tokenizer,
    return_full_text=False,
    max_new_tokens=50,
    do_sample=False,
)
prompt = "Write an email apologizing to Sarah for the tragic gardening mishap"

# Generates the new tokens with pipeline
output = generator(prompt)

# prints the generated text from the first result returned by the pipeline.
print(output[0]['generated_text'])
```

```

The cache for model files in Transformers v4.22.0 has been updated. Migrating
0/0 [00:00<?, ?it/s]

/usr/local/lib/python3.12/dist-packages/huggingface_hub/utils/_auth.py:94: Us
The secret `HF_TOKEN` does not exist in your Colab secrets.
To authenticate with the Hugging Face Hub, create a token in your settings ta
You will be able to reuse this secret in all of your notebooks.
Please note that authentication is recommended but still optional to access p
warnings.warn(
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b3
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b3
Loading checkpoint shards: 100%                2/2 [00:29<00:00, 13.91s/it]

generation_config.json: 100%                181/181 [00:00<00:00, 17.5kB/s]

The `seen_tokens` attribute is deprecated and will be removed in v4.41. Use t
`get_max_cache()` is deprecated for all Cache classes. Use `get_max_cache_sha
WARNING:transformers_modules.microsoft.Phi-3-mini-4k-instruct.f39ac1d28e925b3
Mention the steps you're taking to prevent it in the future.

```

Dear Sarah,

I hope this message finds you well. I am writing to express my sincerest apol

```

# Prompt
prompt = "The capital of France is"

# Tokenize the input prompt as pytorch tensors and take input ids
input_ids = tokenizer(prompt, return_tensors="pt").input_ids

# Token ids sent to gpu to run
input_ids = input_ids.to("cuda")

# model = complete phi-3 model that we loaded earlier ( complete model has 1
# model = Access the internal Transformer network inside Phi-3.
# input ids are sent to the transformer block
model_output = model.model(input_ids)

# The output of the transformer block are sent to lm head
# lm head converts those vectors into token scores.
# model_output[0] = It contains the final representation of every token.last
lm_head_output = model.lm_head(model_output[0])

# Now we have final vector
# 0 = selecting the first sentence , -1 = Last token position
# argmax(-1) = finding the index of the highest score
token_id = lm_head_output[0,-1].argmax(-1)

# the tokenizer has vocabulary with index and words
# Decoding back to text based on index value that mapped to words
tokenizer.decode(token_id)

```

```
'Paris'
```

```
print(model_output[0].shape)
print(lm_head_output.shape)
```

```
torch.Size([1, 5, 3072])
torch.Size([1, 5, 32064])
```

```
prompt = "Write a very long email apologizing to Sarah for the tragic garden"
# Tokenize the input prompt
input_ids = tokenizer(prompt, return_tensors="pt").input_ids
input_ids = input_ids.to("cuda")
```

```
# It measures how long the following cell takes to execute.
%%timeit -n 1
```

```
# Generate the text
generation_output = model.generate(
    input_ids=input_ids,
    max_new_tokens=100,
```

```
# This controls whether the model reuses previously computed Keys and Values
# True = Saves the Keys and Values from previous tokens.
use_cache=True
)
```

The attention mask is not set and cannot be inferred from input because pad t
5.86 s ± 639 ms per loop (mean ± std. dev. of 7 runs, 1 loop each)

```
%%timeit -n 1
```

```
# Generate the text
generation_output = model.generate(
    input_ids=input_ids,
    max_new_tokens=100,
    use_cache=False) # False = # The model doesnt saves the key values of previous
```

32.6 s ± 1.02 s per loop (mean ± std. dev. of 7 runs, 1 loop each)

```
# Using True can be faster for generation
```